

Summary

Ph.D. student in Computer Science at UT San Antonio (GPA 4.0/4.0) with five years of deep learning research experience spanning computer vision, generative modeling, and multimodal learning. Current research focuses on **diffusion models** and **vision-language integration** for controllable generation. Prior work includes an M.S. in Electrical Engineering on deep-learning-based video segmentation, with additional research in object detection and visual tracking.

Education

University of Texas at San Antonio

San Antonio, TX, USA

Ph.D. in Computer Science, Aug 2025 – Present

GPA: 4.0/4.0

- Research focus: Text-conditioned human motion generation and motion style transfer using diffusion models and vision-language models
- Advisor: Dr. Kevin Desai

Shiraz University

Shiraz, Iran

M.S. in Electrical Engineering, Sep 2019 – Sep 2022

GPA: 18.10/20 (Ranked 3rd in class)

- Thesis: *Background Subtraction Based on Deep Neural Networks* — attention blocks and ASPP modules on EfficientNet/ResNet backbones for moving-object segmentation in video
- Advisor: Dr. Mehran Yazdi

Shiraz University

Shiraz, Iran

B.S. in Electrical Engineering, Sep 2014 – Sep 2019

GPA: 15.24/20

- Thesis: *Visual Finger Recognition Platform for Number Sign Language* — CNN-based hand-posture recognition with text and speech output
- Advisor: Dr. Peyman Setoodeh

Research Interests

- Conditioned human motion generation, 3D pose estimation, temporal sequence modeling
- Diffusion models and deep generative modeling for image, video, and motion synthesis
- Vision-Language Models and multimodal conditioning
- Aerial and remote-sensing image analysis: object detection, semantic segmentation, anomaly detection

Technical Skills

- **Languages:** Python, C/C++, MATLAB
- **Frameworks & Libraries:** PyTorch, HuggingFace (Diffusers, Transformers), TensorFlow, OpenCV
- **Focus Areas:** Diffusion models, vision-language models, generative modeling, semantic segmentation, object detection
- **Tools:** Git, Linux, CUDA, Weights & Biases, L^AT_EX

Research Experience

- **Graduate Research Assistant – UT San Antonio, TX, USA** Aug 2025 – Present
Advisor: Dr. Kevin Desai (*Ph.D. thesis work; manuscript in preparation*)
 - Designing and training diffusion models for text-conditioned 3D human motion generation and motion style transfer on the HumanML3D dataset.
 - Leveraging vision-language representations (CLIP) as text encoders to align generated motion with natural-language prompts.
 - Implementing the full training and inference pipeline in PyTorch on a multi-GPU setup; experiments tracked with Weights & Biases.
- **Research Assistant – Signal and Image Processing Lab, Shiraz University, Iran** Sep 2019 – Sep 2022
 - Developed a background-subtraction model combining EfficientNet/ResNet backbones with attention blocks and an ASPP module, improving segmentation accuracy and inference speed on standard benchmarks.
 - Proposed a multi-domain CNN for visual tracking with an Extreme Learning Machine (ELM) output layer, reducing online-learning time while maintaining accuracy.
- **Earlier research (Shiraz University, 2016–2018)**
 - CNN-based hand-posture recognition for number sign language with text-to-speech output; PCA-based face-recognition security system on Arduino.

Selected Projects

- **Diffusion-Based Motion Generation & Style Transfer on HumanML3D**
Training a diffusion model from scratch with CLIP text conditioning for controllable motion generation. *PyTorch, Diffusers, CLIP.*
- **Wasserstein GAN with Gradient Penalty (WGAN-GP)**
Re-implemented WGAN-GP from scratch; studied training stability, gradient-penalty weighting, and mode-collapse on image generation. *PyTorch.*
- **DeepLabV3+ Semantic Segmentation on Pascal VOC 2012**
Trained and evaluated DeepLabV3+ for multi-class segmentation, with transfer learning from an ImageNet-pretrained backbone. *PyTorch, OpenCV.*
- **Vision Transformer (ViT) on Stanford-40 Actions**
Fine-tuned ViT for human-action classification; analyzed attention maps versus CNN baselines. *PyTorch, HuggingFace.*
- **YOLOv3 Object Detection on Raccoon Dataset**
Trained YOLOv3 end-to-end on a custom annotated dataset; analyzed mAP and inference speed. *PyTorch, Darknet.*
- **Knowledge & Self-Distillation on CIFAR-100**
Implemented teacher-student and self-distillation pipelines; analyzed compression-accuracy trade-offs. *PyTorch.*

Work Experience

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| Bamdad Project Laboratory (BPL), Shiraz University | Shiraz, Iran |
| Instructor — Deep Learning with PyTorch | Mar 2024 – Jun 2024 |
| <ul style="list-style-type: none">– Designed and delivered a hands-on course on PyTorch, CNNs, and modern training workflows for upper-year undergraduates. | |
| Shiraz University | Shiraz, Iran |
| Workshop Instructor — Python Programming | Mar 2024 – Apr 2024 |
| <ul style="list-style-type: none">– Led an intensive Python-from-scratch workshop covering NumPy, Pandas, and Matplotlib. | |

- Built Python optimization (GA, simulated annealing) and data-analysis pipelines on GIS data for urban street-configuration and public-transit planning; results used by Shiraz municipality.

Teaching Experience

Over four years of teaching at Shiraz University across five core undergraduate engineering courses — serving as instructor, course designer, and teaching assistant:

- **Linear Algebra** Spring 2022
Instructor, course designer, and grader; developed and graded MATLAB assignments and problem sets.
- **Signals & Systems** Spring 2018 & 2021
Instructor, course designer, and grader; designed MATLAB-based assignments illustrating continuous- and discrete-time signal analysis.
- **Circuit Theory I & II** Spring 2020 & 2021
Instructor, course designer, and grader; led weekly problem-solving sessions for ~40 students per term.
- **Engineering Mathematics** Fall 2020 & 2021
Course designer and grader; wrote and graded homework covering ODEs, complex analysis, and Fourier series.
- **Electronics I & II Laboratory** Fall 2019 & 2020
Laboratory teaching assistant; guided students through circuit prototyping and measurement experiments.

Selected Coursework

UT San Antonio (Ph.D.):

Artificial Intelligence
Data Science
Cloud Computing
Operating Systems
Algorithms

4/4
4/4
4/4
4/4
4/4

Shiraz University (M.S.):

Deep Learning 19.30/20
Machine Vision 18.00/20
Image Processing 19.03/20
Pattern Recognition 19.00/20
Advanced Communication Theory 17.00/20

Awards, Honors & Certifications

- Ranked 3rd GPA in graduating M.S. class, Shiraz University (2022)
- Outstanding Student Branch Award (55 IEEE Iran universities) and First Place in “Motion Festival,” as Secretary of IEEE Shiraz University Branch (2017–2018)
- Coursera: Fundamentals of Reinforcement Learning (2023); Deep Learning Specialization (2019)

References

- **Dr. Kevin Desai**
Assistant Professor, Department of Computer Science, University of Texas at San Antonio
Email: kevin.desai@utsa.edu
- **Dr. Mehran Yazdi**
Professor, Department of Communications and Electronics, School of Electrical and Computer Engineering, Shiraz University
Email: yazdi@shirazu.ac.ir